

Collective Cognitive Security by Design

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What makes human groups cognitively secure and agentic in AI-rich environments?

1. CORE DISTINCTION

"You can outsource parts of thinking. You cannot outsource understanding."



AI can assist

- summarize
- calculate
- translate
- synthesize
- generate options



Humans must retain

- understanding
- context
- critical evaluation
- judgment
- ownership & responsibility

The vulnerability begins when assistance quietly becomes substitution for human judgment.

2. TWO RELATED VULNERABILITIES



Cognitive substitution

AI replaces cognitive acts humans would otherwise exercise.



Premature AI framing

A shared AI-generated frame organizes collective reasoning before uniquely held human knowledge surfaces.

3. TARGET CONSTRUCT

Collective Cognitive Security (CCS)

The capacity of a group to preserve the conditions under which its members can independently generate, challenge, integrate, own, and act on judgments while interacting with AI.

High task performance does not guarantee human cognitive security.

The missing layer is process integrity: who retrieved, questioned, evaluated, chose, and owned the reasoning?

It asks whether people still:

- | | | | |
|----------|-----------|----------|-----------|
| generate | challenge | evaluate | integrate |
| choose | own | act | |

4. FOUR DIFFERENT QUESTIONS



Psychological Safety

Can I speak?



Epistemic Agency

Can I generate, judge, and own?



Collective Intelligence

How well can the group perform?



Collective Cognitive Security

Can humans remain epistemic actors while AI participates?

5. THE EVIDENCE GAP

Human + AI is not automatically the best system

Augmentation is not the same as synergy.

Vaccaro, Almaatouq & Malone (2024): across 106 experiments, Human+AI often improved over humans alone, but did not on average outperform the best available component.

Interaction architecture matters

7. WHAT TO MEASURE

- uniquely held human knowledge surfaced
- enacted epistemic participation
- cognitive diversity
- decision ownership
- responsibility
- idea provenance
- AI reliance
- cognitive-capability retention

6. DESIGN HYPOTHESIS

AI timing and interaction architecture are cognitive-security design variables



Protected Human Divergence
candidate design principle



Designed Collective Thinking (DCT)
candidate social architecture



Epistemic Agency Practice Loop (EAPL)
candidate mechanism / process model

Independent human retrieval



Protected human divergence



Structured human deliberation



AI augmentation



Human re-evaluation



Human ownership and action

Human First does not mean Human Only.

This is a testable candidate architecture, not a prescription.

8. COGNITIVE VULNERABILITY & MITIGATION CATALOG



Vulnerability: premature AI framing and cognitive substitution can reduce human epistemic participation and prevent uniquely held human knowledge from entering collective reasoning.



Mitigation / governance hook: protect a phase of human generation before consequential shared AI framing.



Metric: uniquely held human knowledge surfaced + enacted epistemic participation.

Not only humans in the loop. Humans developing in the loop.